

PHYSICS

1. The moment of inertia of a uniform rod of mass m and length l is α when rotated about an axis passing through centre and perpendicular to the length. If the rod is broken into equal halves and arranged as shown, then the moment of inertia about the given axis is

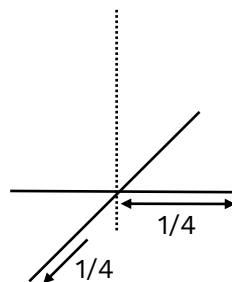
- (1) 2α
 (2) $\alpha/2$
 (3) 4α
 (4) $\alpha/4$

Sol. $I = \frac{mL^2}{12} = \alpha$

$$I_1 = \left[\frac{\frac{M}{2} \left(\frac{L}{2} \right)^2}{12} \right] 2$$

$$I_1 = \frac{1}{4} \frac{ML^2}{12} = \frac{\alpha}{4}$$

2. Match the following



- (a) Viscosity $\rightarrow ML^{-1}T^{-1}$
 (b) Compressibility $\rightarrow M^{-1}L^1T^2$
 (c) Pressure gradient $\rightarrow M^1L^{-2}T^{-2}$
 (d) Intensity of wave $\rightarrow M^1L^0T^{-3}$

Sol. By $F = \eta A \frac{dv}{dx}$: dimension viscosity

$$[\eta] = [ML^{-1}T^{-1}]$$

$$\text{Compressibility } K = \frac{1}{B(\text{Bulk modulus})}$$

$$[K] = [M^{-1}LT^{-2}]$$

$$\text{Pressure gradient} = \frac{\text{pressure}}{\text{length}} = [ML^{-2}T^{-2}]$$

$$\text{Intensity of wave} = \frac{\text{Power}}{\text{Area}} = [ML^0T^{-3}]$$

3. Which of the following statement(s) is correct is/are for the adiabatic process?

- A. Molar heat capacity is zero.
 B. Molar heat capacity is infinite.
 C. Work done on gas is equal to increase in internal energy
 D. The increase in temperature results in decrease in internal energy

- (1) A and C only
 (2) B and C only
 (3) A and D only
 (4) C and D only

Sol. $C_m = \frac{q}{n\Delta T}$

For adiabatic process $q = 0$, $C_m = 0$

$$\Delta U = q + w \Rightarrow \Delta U = w$$

work done on gas = +ve, $\Delta U = +ve$

$T \uparrow \Delta U \uparrow$, so option (1) is correct.

- 4.** Two point charges q and $9q$ are placed at distance of l from each other. Then the electric field is zero at a

- (1) Distance $\frac{l}{4}$ from charge $9q$
 (2) Distance $\frac{3l}{4}$ from charge q
 (3) Distance $\frac{l}{3}$ from charge $9q$
 (4) Distance $\frac{l}{4}$ from charge q

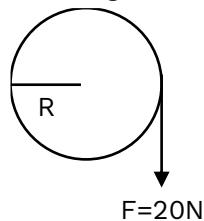
Sol. $\frac{Kq_1}{x^2} = \frac{K(q_2)}{(\ell - x)^2} \Rightarrow \frac{kq}{x^2} = \frac{K[9q]}{[\ell - x]^2}$

$$\ell - x = 3x$$

$$\ell = 4x$$

$$x = \frac{\ell}{4}$$

- 5.** $m = 10$ kg, $R = 10$ cm. What is ω when the string is unwound by 1 meter.



Sol. Assume Disc

$$F(R) = I\alpha$$

$$RF = \frac{MR^2}{2} \times \alpha$$

$$\ell = R\theta$$

$$1 = 0.1 \times \theta$$

$$\theta = 10 \text{ rad}$$

$$\alpha = \frac{2F}{MR}$$

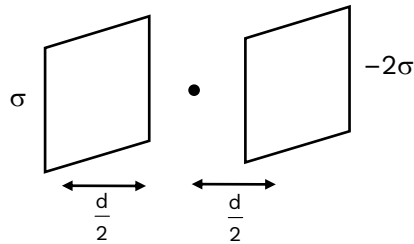
$$\alpha = \frac{2 \times 20}{10 \times 0.1}$$

$$\alpha = 40 \text{ rad/sec}^2 \rightarrow \omega_f^2 = \omega_i^2 + 2\alpha\theta$$

$$\omega^2 = 20 \times 40$$

$$\omega = 20\sqrt{2} \text{ rad/sec}$$

6. Calculate the force on the charge Q placed between the two infinite sheets of surface charged density $+\sigma$ and -2σ are the charge density on the plates.



Sol.
$$\left(\frac{\sigma + 2\sigma}{2\epsilon_0} \right) Q = \frac{3}{2} \frac{\sigma Q}{\epsilon_0}$$

7. The ratio of magnetic field to center of circular coil to magnetic field at distance x from the centre of circular coil $\left(\frac{x}{R} = 3/4 \right)$.

Sol.
$$\frac{B_1}{B_2} = \frac{\frac{\mu_0 i}{2R}}{\frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}}$$

$$x = \frac{3}{4} R$$

$$\frac{B_1}{B_2} = \frac{64}{125}$$

8. γ_A : monoatomic gas
 γ_B : Polyatomic [consider 1 vibrational mode]

Find $\frac{\gamma_A}{\gamma_B}$.

Sol. $f = 3T + 3R + 2$
 $f_B = 8$ (Assume non linear)

$$\frac{\gamma_A}{\gamma_B} = \frac{4}{3}$$

$$\gamma_A = 1 + \frac{2}{\delta_A} = \frac{1+2}{3} = \frac{5}{3}$$

$$\gamma_B = \frac{1+2}{f_B} = \frac{1+2}{8} = \frac{5}{4}$$

9. A current carrying wire is in form of a circle of radius R . Find ratio of magnetic field at centre to the magnetic field at axial point at a distance R from its centre.

10. $x + y + z = K$, is the equation of wavefront, find the angle made by direction of propagation of wave with x axis.

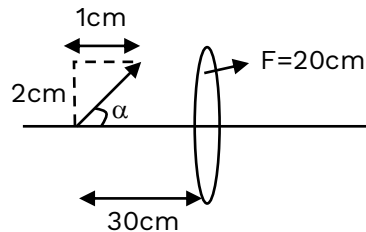
Sol. Normal vector = $\hat{i} + \hat{j} + \hat{k}$

When

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot \hat{i}}{\sqrt{3}(1)}$$

$$\cos \theta = \frac{1}{\sqrt{3}}$$

11. Find the angle made by the image with the principal axis.



Sol. $u = -30$

$f = +20$

$v = ?$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$$

$V = 60 \text{ cm}$

$$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{30}$$

$$\frac{h_i}{h_o} = \frac{v}{u} \quad \tan B = \frac{4}{4}$$

$$h_i = 4 \text{ cm} \quad B = 45^\circ$$

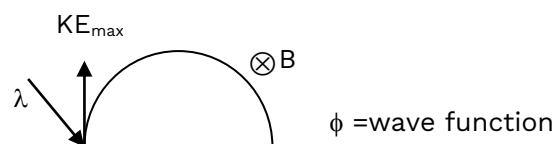
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$-\frac{1}{v^2} dv - \left(-\frac{1}{4^2} du \right) = 0$$

$$dv = \frac{v^2}{u^2} \times du$$

$$= 4 \times 1 = 4 \text{ cm}$$

12. An electron is ejected normal to the surface then the diameter of the circular path is:



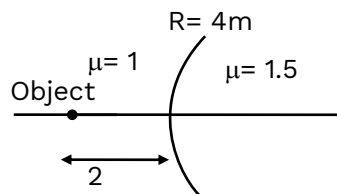
Sol. $KE_{\max} = \frac{hc}{\lambda} - \phi$

$$\frac{1}{2} m u^2 = \frac{\mu c}{\lambda} - \phi$$

$$v = \sqrt{\frac{2}{m} \left(\frac{h c}{\lambda} - \phi \right)}$$

$$d = 2R = 2 \left[\frac{m v}{2B} \right] = \frac{m}{B} \sqrt{\frac{2}{m} \left(\frac{h c}{\lambda} - \phi \right)}$$

13. Find the position of image



Sol. $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

$$u = -2 \text{ m}$$

$$n_2 = 1.5$$

$$R = +4 \text{ m}$$

$$n_1 = 1$$

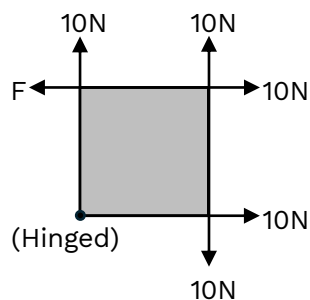
$$\frac{3}{2v} - \frac{1}{-2} = \frac{1}{4}$$

$$\frac{3}{2v} + \frac{1}{2} = \frac{1}{4}$$

$$\frac{3}{v} = \frac{1}{8} - \frac{1}{2}$$

$$\frac{3}{2v} = \frac{-3}{8} \Rightarrow v = -4 \text{ m}$$

14. A square shape lamina of mass M kg is at rest. Find value of F (in N).



(1) 10 N

(2) 15 N

(3) 20 N

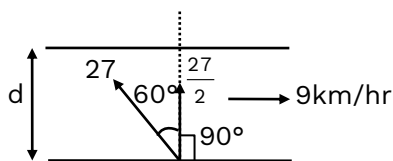
(4) 30 N

Sol. $F \ell = 10 \times \ell$

$$F = 10 \text{ N}$$

15. A boat can move with 27 km/h in still water. If the river is flowing with speed of 9 km/h, and the boat is moving at an angle of 150° from the flow of river and takes 30 sec to cross the river. Find the width of the river.

Sol. $d = \frac{27}{2} \times \frac{5}{18} \times 30 = 112.5 \text{ m}$



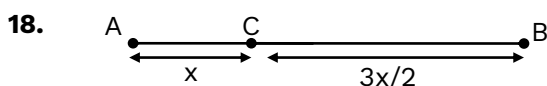
16. A two SHMs are superimposed on a particle moving along x axis as $x_1 = A \sin \omega t$ and $x_2 = A \sin(\omega t + \pi/3)$ then acceleration of particle is (given $\omega=5$)

Sol. $x_{\text{net}} = \sqrt{3}A \sin(\omega t + \frac{\pi}{6})$

$$a = A\omega^2 = (\sqrt{3}A)[5]^2 = 25\sqrt{3}A$$

17. What is the ratio of radius of n^{th} orbit in H, He^+ and Li^{+2} ? (Assume Bohr model is applicable)

Sol. $r = r_0 \times \frac{n^2}{z}$



If speed in part AC is 5 m/sec, and average speed in part AB is $50/7$ m/sec, then find speed in part CB.

Sol.
$$\frac{1 + \frac{3x}{2}}{\frac{3x}{2}} = \frac{50}{7}$$

$$\frac{x}{5} + \frac{2}{v_{BC}} = \frac{5}{7}$$

$$x \frac{\frac{5x}{2}}{\left[\frac{1}{5} + \frac{3}{2v} \right]} = \frac{5}{7}$$

$$7 = 20 \left(\frac{1}{5} + \frac{3}{2v} \right)$$

$$7 = 4 + \frac{60}{2v}$$

$$3 = \frac{30}{v}$$

$$V = 10$$

19. Choose the correct relation between relative permeability and magnetic susceptibility.

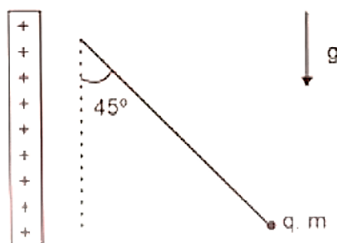
Sol. $\mu = \mu_0 \mu_r$
 $\mu = \mu_0 [1 + x_m]$
 $\mu_r = 1 + X_m$

- 20.** Find the dimension of $\frac{a}{b^2}$ from the given formula $\left(P + \frac{a}{V^2}\right)(V - b) = RT$ where symbols have their usual meaning.

- (1) ML^3T^{-2}
 (2) $ML^{-1}T^{-2}$
 (3) ML^3T^{-1}
 (4) MLT

Sol. $P = \frac{a}{V^2}$
 $a = PV^2$
 $b = V$
 $\frac{PV^2}{V^2} = P$

- 21.** The figure shows an infinite plane having uniform charge density σ and a small charged particle having charge q and mass m suspended by a light insulating thread. Find σ if the charge is in equilibrium.



Sol. $qE = mg$
 $q = \left(\frac{\sigma}{2\epsilon_0}\right) = mg$
 $\sigma = \frac{2\epsilon_0 mg}{q}$